INCH-POUND

A-A-60062

May 9, 2024

COMMERCIAL ITEM DESCRIPTION

STUDS, CADMIUM AND HEXAVALENT CHROMIUM FREE

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1 SCOPE. This commercial item description (CID) covers studs containing no cadmium or hexavalent-chromium in their finishes.

1.1 Intended use. Studs contained herein are intended for ground vehicle applications. Studs described in this CID shall not be used for aerospace applications without design authority approval.

1.2 Clean finish. See Section 7 (paragraph 7.3) for definition of clean finish when referenced on a table.

2 CLASSIFICATION

2.1 Types.

Type I	-	AIA/NAS184 - Stud, Fine Thread
Type II	-	AIA/NASM51864 - Stud, Plain, General Purpose

3 SALIENT CHARACTERISTICS

3.1 Materials. Unless otherwise specified herein, the studs shall be manufactured as specified in the Aerospace Industry Association (AIA) documents listed above by type with the exception of, the plating/finishes shall be free of cadmium and hexavalent chromium finishes as specified herein. The use of recovered materials made in compliance with regulatory requirements is acceptable provided all requirements of this CID are met.

3.2 Dimensions. Dimensions for all designated types of studs apply after plating/finishing and prior to any application of lubricants (if any). Refer to each individual NAS/NASM specification listed in Section 2.1 for detailed part geometry and tolerance requirements.

3.3 Hexavalent-chromium free Part or Identifying Number (PIN) codes and finish/plating options. PIN finish/plating codes for all designated types of studs:

QX	=	Carbon/alloy steel shall be zinc nickel plated in accordance MIL-PRF-32647, Type K16.
P5	=	CRES shall be cleaned, descaled and passivated in accordance with SAE AMS2700, Method 1, Type 6, 7, or 8, or Method 2. Hexavalent chromium containing compounds shall be prohibited.

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of the address information using the ASSIST Online database at https://quicksearch.dla.mil.

FSC 5307

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3.4 Lubrication. Lubrication types (if any) for each type of stud shall be specified in each AIA/NASM specification (see 2.1).

3.5 Interchangeability. When AIA/NASM specifications list interchangeability parts, do not interchange parts containing cadmium or hexavalent chromium with the cadmium and hexavalent chromium-free parts listed herein without design authority approval.

4 REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered, recycled, bio based, and environmentally preferable materials to the maximum extent practicable, in accordance with paragraph 23.403, 23.404(b) of the Federal Acquisition Regulation (FAR), or the sustainable acquisition of EO 14057; as applicable.

5 PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market.

5.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.3 Contractor certification. The contractor shall certify and maintain substantiating evidence the product offered meets the salient characteristics of this commercial item description and conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to perform or witness any of the inspections it deems necessary to ensure supplies and services conform to prescribed requirements.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7 NOTES

The following part or identification numbering (PIN) procedure is for government purposes and does not constitute a requirement for the contractor.

7.1 Tables. The following finish/plating PIN's are cadmium and hexavalent chromium-free for designated stud types.

TABLE I. Type I (-1) stud, fine thread.

(Use NASM51864 for material, sizes/lengths, and heat treatment 125 to 145 KSI per AMS2759.)

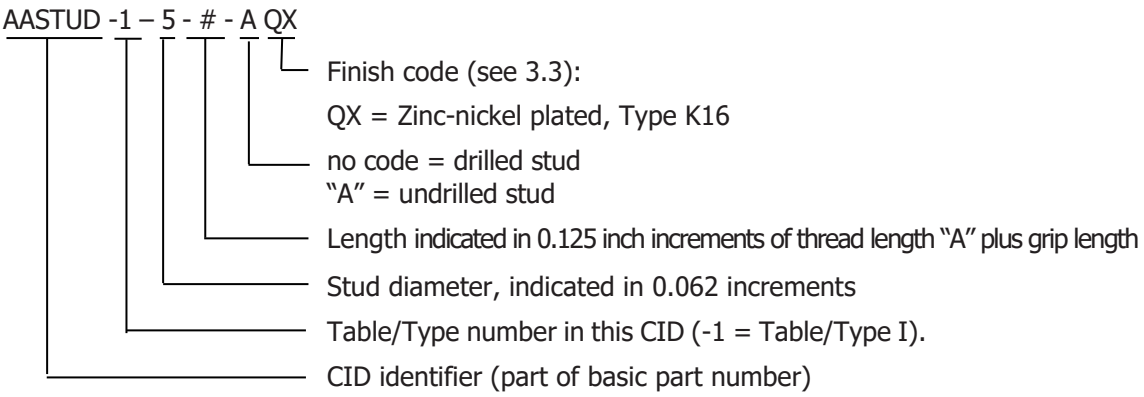
A-A-STUD Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NAS184 Referenced "E Thread" Cadmium Plated, DO NOT USE on Ground Vehicles	NAS184 Stud Diameter Dash Number
AASTUD-1-3-#*	0.1900-32	3
AASTUD-1-4-#*	0.2500-28	4
AASTUD-1-5-#*	0.3125-24	5
AASTUD-1-6-#*	0.3750-24	6
AASTUD-1-7-#*	0.4375-20	7
AASTUD-1-8-#*	0.5000-20	8

NOTES:

Part number for clean finished studs in this CID shall be as described in the "PIN construction" (next page).

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PIN construction:



Examples of clean finish PIN's:

AASTUD-1-5-23QX = a NAS184 – stud, Ø.312, length 2.875, drilled hole in stud, zinc nickel plated, Type K16.
AASTUD-1-5-23AQX = a NAS184 – stud, Ø.312, length 2.875, undrilled stud, zinc nickel plated, Type K16.

TABLE II. Type II (-2) stud, plain, general purpose.

(Use NASM51864 for material, heat treatment (if any), and sizes/lengths.)

A-A-STUD Part Number Zinc-Nickel Type K16 Plated (Clean Finish)	NASM51864 Reference Part Number Cadmium Plated, DO NOT USE on Ground Vehicles	A-A-STUD Part Number Passivated (CRES) (Clean Finish)	NASM51864 Reference Part Number Passivated (CRES), DO NOT USE on Ground Vehicles
AASTUD-2-102-#*	MS51864-102#*	AASTUD-2-202-#*	MS51864-202#*
AASTUD-2-103-#*	MS51864-103#*	AASTUD-2-203-#*	MS51864-203#*
AASTUD-2-104-#*	MS51864-104#*	AASTUD-2-204-#*	MS51864-204#*
AASTUD-2-105-#*	MS51864-105#*	AASTUD-2-205-#*	MS51864-205#*
AASTUD-2-106-#*	MS51864-106#*	AASTUD-2-206-#*	MS51864-206#*
AASTUD-2-107-#*	MS51864-107#*	AASTUD-2-207-#*	MS51864-207#*
AASTUD-2-108-#*	MS51864-108#*	AASTUD-2-208-#*	MS51864-208#*
AASTUD-2-109-#*	MS51864-109#*	AASTUD-2-209-#*	MS51864-209#*
AASTUD-2-110-#*	MS51864-110#*	AASTUD-2-210-#*	MS51864-210#*
AASTUD-2-111-#*	MS51864-111#*	AASTUD-2-211-#*	MS51864-211#*
AASTUD-2-112-#*	MS51864-112#*	AASTUD-2-212-#*	MS51864-212#*
AASTUD-2-113-#*	MS51864-113#*	AASTUD-2-213-#*	MS51864-213#*

NOTES:

(Brass and nickel-copper alloy per NASM51864 do not apply to this CID.)

Material, non-CRES per AIR4127; CRES per AIR4127, 300 series; mechanical properties per FF-S-1362.

Part number for clean finished studs in this CID shall be as described in the "PIN construction" (see next page).

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PIN construction:

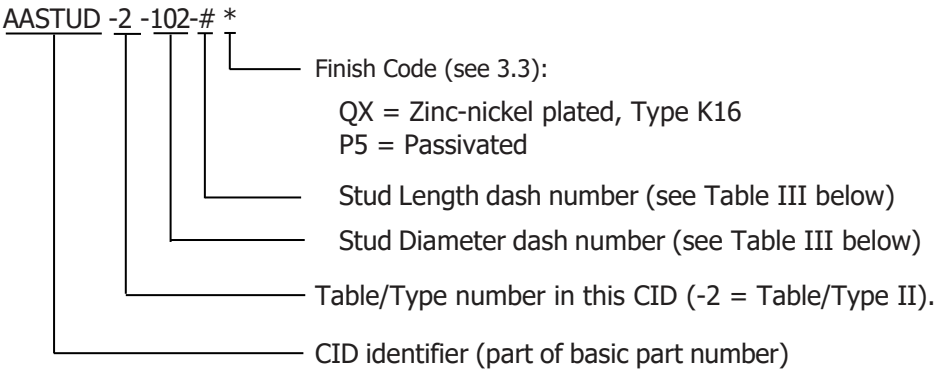


Table III. Length and diameter dash numbers.

Length Dash No.	Diameter Dash Number from NASM51864											
	102	103	104	105	106	107	108	109	110	111	112	113
	202	203	204	205	206	207	208	209	210	211	212	213
	Overall Stud length											
12	1.500	--	--	--	--	--	--	--	--	--	--	--
14	1.750	1.750	--	--	--	--	--	--	--	--	--	--
16	2.000	2.000	2.000	--	--	--	--	--	--	--	--	--
18	2.250	2.250	2.250	2.250	--	--	--	--	--	--	--	--
20	2.500	2.500	2.500	2.500	2.500	--	--	--	--	--	--	--
22	2.750	2.750	2.750	2.750	2.750	2.750	--	--	--	--	--	--
24	3.000	3.000	3.000	3.000	3.000	3.000	3.000	--	--	--	--	--
26	--	--	3.250	3.250	3.250	3.250	3.250	--	--	--	--	--
28	--	--	3.500	3.500	3.500	3.500	3.500	3.500	--	--	--	--
30	--	--	3.750	3.750	3.750	3.750	3.750	3.750	--	--	--	--
32	--	--	4.000	4.000	4.000	4.000	4.000	4.000	4.000	--	--	--
34	--	--	--	--	4.250	4.250	4.250	4.250	4.250	--	--	--
36	--	--	--	--	4.500	4.500	4.500	4.500	4.500	4.500	--	--
38	--	--	--	--	4.750	4.750	4.750	4.750	4.750	4.750	4.750	--
40	--	--	--	--	5.000	5.000	5.000	5.000	5.000	5.000	5.000	--
44	--	--	--	--	--	--	5.500	5.500	5.500	5.500	5.500	5.500
48	--	--	--	--	--	--	6.000	6.000	6.000	6.000	6.000	6.000
52	--	--	--	--	--	--	--	--	--	--	6.500	6.500
56	--	--	--	--	--	--	--	--	--	--	7.000	7.000
60	--	--	--	--	--	--	--	--	--	--	7.500	7.500
64	--	--	--	--	--	--	--	--	--	--	8.000	8.000

Examples of clean finish PIN's:

AASTUD-2-102-16QX = a NASM51864 – stud, carbon or alloy steel, Ø.2500 inches, 2 inches length, zinc nickel Type K16 plated.
AASTUD-2-204-16P5 = a NASM51864 – stud, CRES, Ø.3750 inches, 2 inches length, passivated.

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7.2 Source of documents.

7.2.1 FAR: Copies of the Federal Acquisition Regulations are available online from https://www.acquisition.gov/.

7.2.2 Executive order (EO). Copies of EO 14057 are available online from https://www.fedcenter.gov/programs/eo14057/.

7.2.3 Military specifications. Copies of military specifications are available online from https://quicksearch.dla.mil.

7.2.4 AIA documents. Copies of Aerospace Industries Association documents are available online from https://www.aia-aerospace.org/.

7.2.5 SAE documents. Copies of SAE International documents are available online from https://www.sae.org/.

7.3 Clean finish definition. In the tables, where it says, "clean finish", clean finish is defined as treatments which do not include more than 0.1 percent by weight Cadmium or Hexavalent Chromium in ***any*** homogenous layer.

7.4 Ordering data. The contract or order should specify the following:

a. Title, number and date of this CID.
b. National stock number (if available).
c. Material.
d. Hexavalent chromium or cadmium free part number (PIN) from this CID.
e. Quantity.

7.5 Key words:

Alloy steel
Carbon steel
Clean finish
CRES
Plate
Plating
Steel
Zinc-nickel

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CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – AT
Navy – SH
Air Force – 71
DLA – CC

Review activities:
Army – AR, AV, CR, MI
Navy – CG, MC, YD
Air Force – 03, 70
DLA – GS4, GS7

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing Activity:
Army – AT

(Project 5307-2024-003)

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